SIX Index IT Home Assignment: Index Reviewer App for SMI

Context

Design and implement an **Index Reviewer Application** in **Python or Java**.

The application shall support the execution of an **SMI Index Review for Q3 2026** and expose its functionality through an API.

The objective is not only to produce the correct review outcome, but also to demonstrate sound software engineering practices, architectural thinking, extensibility, and technical decision-making.

The application should be designed with future evolution in mind, allowing support for additional indices, review schedules, and business requirements.

Instructions

General Expectations

- You may use AI-assisted development tools; however, you must be able to explain, justify, and maintain all submitted code.
- Open-source libraries and frameworks may be used.
- Follow-up technical discussions will be conducted based on your submission.
- Input data are provided as CSV files containing:
    - SPI universe data (spi_universe.csv)
    - Security reference data (sec_data.csv)
    - Current SMI composition (composition.csv)
- Where requirements are ambiguous, make reasonable assumptions and document them clearly.
- If a requirement is unclear, questions may be sent to: lucas.damalix@six-group.com and sorin.ivascu@six-group.com

Deliverable

Submit a ZIP archive containing at least:

- Complete source code
- Build and execution instructions

Timing

The assignment should require approximately **6-8 hours** of work.

Submission deadline: 72 hours after receipt

Review Process

The review should follow the principles described in the **SIX methodology rulebook (section 5.12.3)**. The objective is to determine the SMI composition for **Q3 2026** using the provided market and security data.

The review process is based on two dates:

- **Cut-off Date:** used to evaluate prices for weight calculation.
- **Review Date:** used to evaluate all other data required for the review.

For Q3 2026: cut-off = 2026-09-10, review date = 2026-09-21

Eligible securities (securities from the SPI universe) are ranked using **Free Float Market Capitalization (FFMCAP)**.

Free float market capitalization is calculated using this formula:

> *FFMCAP = Price(t') × Shares(t) × Free Float(t)*
>
> *where t' is the cut-off date and t the review date*

Based on these rankings, the application shall determine the constituents qualifying for inclusion in the SMI.

The index should be constructed from the **20 highest-ranked eligible securities**. A selection buffer may be considered (explained in section 5.12.3.2 of the rulebook).

Once the constituents have been selected, their index weights shall be calculated and adjusted (using capping factors) so that no constituent exceeds the maximum weight permitted by the methodology (**18% cap**).

The exact implementation details are intentionally left open. You are expected to make reasonable assumptions where necessary, document them clearly, and justify their design decisions.

Simplified Example

The example below illustrates the review workflow with a simplified universe of 5 companies and an index containing 3 constituents. The constituent cap is set to **50%** for illustration purposes.

1. Rank the Eligible Companies

Eligible securities are ranked according to the methodology criteria (e.g. Free Float Market Capitalization).

Company	Price	Shares	Free Float	FFMCAP	FFMCAP Rank
A	3	200	1	600	1
B	6	50	1	300	2
C	1	200	0.5	100	3
D	10	5	1	50	4
E	0.5	20	1	10	5

The top 3 ranked companies are selected: A, B and C. D and E are not.

2. Calculate Raw Weights

The selected constituents have the following Free Float Market Capitalizations:

Company	FFMCAP
A	600
B	300
C	100
Total	**1000**

Raw weights:

Company	Weight
A	60%
B	30%
C	10%

3. Apply the 50% Cap

No constituent may exceed 50%.

Company A exceeds the cap:

Company	Raw Weight	Capped Weight
A	60%	50%
B	30%	30%
C	10%	10%

After capping:

- Total allocated weight = 90%
- Excess weight = 10%

4. Redistribute the Excess Weight

The excess weight is redistributed among constituents that are still below the cap.

Current eligible constituents:

- B (30%)
- C (10%)

Redistribution ratio:

- B receives 30 / (30 + 10) = 75%

- C receives 10 / (30 + 10) = 25%

Redistributed weight:

- B receives 7.5%
- C receives 2.5%

Final weights:

Company	Final Weight
A	50%
B	37.5%
C	12.5%

All constituents are now below the cap and the weights sum to 100%.

Requirements

Functional

The application shall:

- Calculate the **new composition of the SMI for Q3 2026**.
- Calculate the **weighting factors** of all resulting SMI constituents.
- Produce a review report containing:
  - Final constituent list
  - Weighting factors
  - Joiners
  - Leavers
  - Review status

Non-functional

- Data validation
- Traceability
- Design documentation

Nice to have

Additional enhancements may be included, but evaluation will prioritize simplicity, clarity, maintainability, and sound design over feature quantity.

- Be designed in a manner that supports:
  - Additional indices
  - Additional review dates
  - New review rules
  - Future enhancements with minimal refactoring
- Automated Testing
- Auditability
- Extensibility

Out-of-scope

- User management
- User interface
- Infrastructure provisioning
- CI/CD
- Production deployment concerns